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Explainable AI-Based Student Performance and Dropout Risk Prediction for Early Identification of At-Risk Students

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Abstract—Now, huge amounts of educational data are generated by the learning management system, attendance, and continuous assessment process, thereby enabling the identification of students who are likely to encounter difficulties or drop out well before performance declines. Conventional monitoring practices, which are largely based on performance assessment and teacher's observations, are inadequate in addressing the multiple variables and complex interactions that determine a student's progress. Machine Learning (ML) has proven to be a reliable solution for recognizing patterns within students' records, but these patterns are sometimes difficult to interpret, thus rendering them ineffective. This paper proposes an Explainable AI-based framework for predicting student performance and identifying students at academic risk while providing transparency by interpreting the reasons behind the predictions using SHAP (SHapley Additive exPlanations). The framework uses academic records, attendance, behavior, family, and social features extracted from the UCI Student Performance dataset to predict student performance and academic risk. It is designed using the final grade (G3) as a performance metric and developing an alternative indicator for the absence of a dropout label. The proposed methodology involves data preprocessing, feature engineering and selection, performance and at-risk prediction using Random Forest and XGBoost, SHAP-based model interpretation, and a rule-based personalized intervention module, which transforms identified risk factors into interpretable insights for the end users. By prioritizing prediction transparency and providing relevant recommendations for intervention, the framework is expected to become more than just a prediction model but a trusted advisor to help teachers intervene with at-risk students before it is too late.

Keywords—Predicting student performance; predicting dropout risk; machine learning; explainable AI; SHAP; educational data mining; early intervention; student analytics.

I. INTRODUCTION

The rapid adoption of learning management systems has led to the generation of a large amount of student-related data. Currently, educational institutions collect student records, which contain information on their academic performance, attendance, behavior, study habits, family details, and level of engagement in learning activities. This volume and variety of data can be leveraged using Artificial Intelligence (AI) and Machine Learning (ML) to monitor academic performance and intervene early with students who are at academic risk or whose performance is below average.

A student's performance is a function of multiple variables, some of which include previous grades and scores, number of hours spent studying, school attendance record, family's

academic encouragement, educational background, and social circle among others. Traditional performance monitoring practices are largely based on performance assessment and teacher's observation, which are primarily subjective measures. These approaches are mostly ineffective in addressing the complex interactions and many variables that contribute to a student's performance. With the ability to identify patterns within students' records, Machine Learning (ML) has proven to be a reliable solution for predicting performance and identifying students at academic risk. However, these predictions can be difficult to interpret, thereby rendering them ineffective. Explainable Artificial Intelligence (XAI) addresses this issue by providing a human understandable explanation of the underlying factors that lead to a certain prediction. In this context, SHAP can be used to determine the contribution of each feature to a given prediction, thus explaining what causes a student's performance to decline or improve.

This project involves developing an innovative approach comprising of performance prediction, at-risk prediction, explainable AI, and personalized recommendation modules into a single framework that can be trusted to support early intervention decisions. It aims to combine attendance, behavior, socio-economic, and academic data to provide explanations for performance predictions as well as recommendations for improving performance and addressing the causes of performance decline. The UCI Student Performance dataset will be used as the primary source of student records on which the framework will be trained and evaluated. The dataset comprises of 395 instances with 33 attributes that contain information on a student's demographics, family, schools, social life, lifestyle, and academic performance. The objective will be to predict final grade (G3), which is the final term Mathematics grade, using other attributes. A limitation of this dataset is that it does not include a class label for students who dropped out of school or were withdrawn by their parents. Instead, an alternative approach will be used to determine whether a student is at academic risk or not based on their number of failures prior to the final term, attendance record, and other attributes that indicate such risks. The proposed solution involves developing a data preprocessing and feature engineering pipeline, conducting feature selection, building performance and at-risk prediction models using Random Forest and XGBoost, using SHAP to interpret model results, and developing a recommendation engine that can recommend interventions based on the risk factors identified by the model. Ultimately, the proposed framework will support effective early intervention by providing teachers with reasons why a student's performance is likely to decline and what they can do to prevent it.

II. LITERATURE SURVEY

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Cortez and Silva [1] investigated the prediction of secondary student performance using data mining techniques. Their work demonstrated that many attributes, including demographic, social, and school-related information, had significant predictive power and made a valuable contribution by providing a dataset and methodology [2] that is the basis of this project as well as much of the following work on educational data mining. Several works have since focused on student performance prediction with multiple approaches including classifier systems, regression models, and clustering algorithms. These works have demonstrated that, compared to traditional linear approaches, ensemble methods are better at modeling the complex, non-linear interactions between input attributes and the target variable. Breiman [4] proposed the Random Forest algorithm, which makes use of bagging and boosting to improve the performance of decision trees and provide feature importances, thereby enabling good predictions while remaining interpretable. Chen and Guestrin [5] proposed the XGBoost algorithm, which facilitates faster and more accurate model training, compared to many other tree-based approaches, thus becoming a popular choice for structured data such as the Student Performance dataset. Pedregosa et al. [6] published the Scikit-learn library, which has since become the standard for implementing and evaluating many of these algorithms in practice.

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The algorithms identified above have proven to be fairly successful at making accurate performance predictions. However, interpreting the results of such models is often non-trivial, especially when it comes to understanding which attributes contribute most to a given prediction. According to Lundberg and Lee [3], SHAP offers a unified approach to explain the output of a model by relying on a game-theoretic framework to estimate the contribution of each attribute to a prediction. Several surveys, including Molnar's [7] and Geron's [8], describe SHAP values as one of the key approaches to model interpretation. Indeed, SHAP, along with feature importances and partial dependence plots, has become an essential requirement for any Machine Learning algorithm, particularly in domains such as education and health care, in which decisions must be interpretable.

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Many researchers have worked on the identification of students at risk of academic failure or dropping out of school. Most of these works have focused on the utilization of academic records, attendance data, demographics, socio-economic status, and behavioral patterns to predict such events. However, while many of these works have demonstrated the accuracy of their approaches, the interpretation of the results remains a challenge. Prenkaj et al. [12] provided an extensive survey of many Machine Learning methods used to predict student dropout in online courses. Lykourantzou et al. [14] and Marquez-Vera et al. [15] demonstrated the effectiveness of classical classification algorithms in early warning systems, while Kruger et al. [11] and Vaarma and Li [13] introduced explanations for such models. These works demonstrate that, while classical classification algorithms are effective at predicting student failure, they are difficult to interpret, thus limiting their applicability in practice.

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Similarly, while feature selection and engineering can facilitate interpretation to an extent, they are often insufficient, particularly when it comes to complex, non-linear pattern recognition algorithms such as Random Forest or XGBoost. Finally, most works only focus on the prediction of an event, such as student dropout, without further exploring how the identified factors can be utilized to improve students' performance.

A comprehensive review of the field can be found in the works of Romero and Ventura [9] and Baker [10], which highlight the importance of educational data mining in general, as well as the unique challenges associated with student performance prediction in particular. On the whole, the works discussed above serve as the foundation for the current one by highlighting the areas that require further research, such as the combination of different aspects of students' records, the utilization of explainable ML models, and the inclusion of personalized recommendations and interventions.

III. PROPOSED WORK

The aim of this project is to propose a student analytics framework based on Explainable AI that can be used to predict academic performance and identify students at risk of dropping in order to facilitate early intervention. The framework will rely on academic performance records, attendance data, behavior, socio-economic factors, and social information. It will provide a performance prediction, determine whether a student is at risk and identify the factors that contribute to either of these outcomes using SHAP, and provide intervention recommendations based on those factors. Overall, it will rely on these key components:

1. Data preprocessing and feature engineering and selection to prepare students' records for use as input to the performance prediction and at-risk detection models.
2. Performance prediction and at-risk detection models, which will rely on Random Forest and XGBoost to provide fairly accurate predictions of student performance as well as detect those students who are likely to encounter academic problems.
3. An explainable AI component that will utilize SHAP values to identify the factors that contribute to a model's prediction and provide a human comprehensible interpretation of the results.
4. A recommendation engine that will serve as an intervention advisor by recommending individualized, data-driven recommendations based on the factors a model identifies as important.

Ultimately, by providing accurate predictions of students' performance and academic risk and interpreting these results using SHAP, the framework will become more than just a prediction model. It will provide teachers with valuable, individualized recommendations that can help them take early intervention measures to ensure that students who are unlikely to succeed without assistance get the support they need. The framework will be presented using the architecture in Fig. 1.

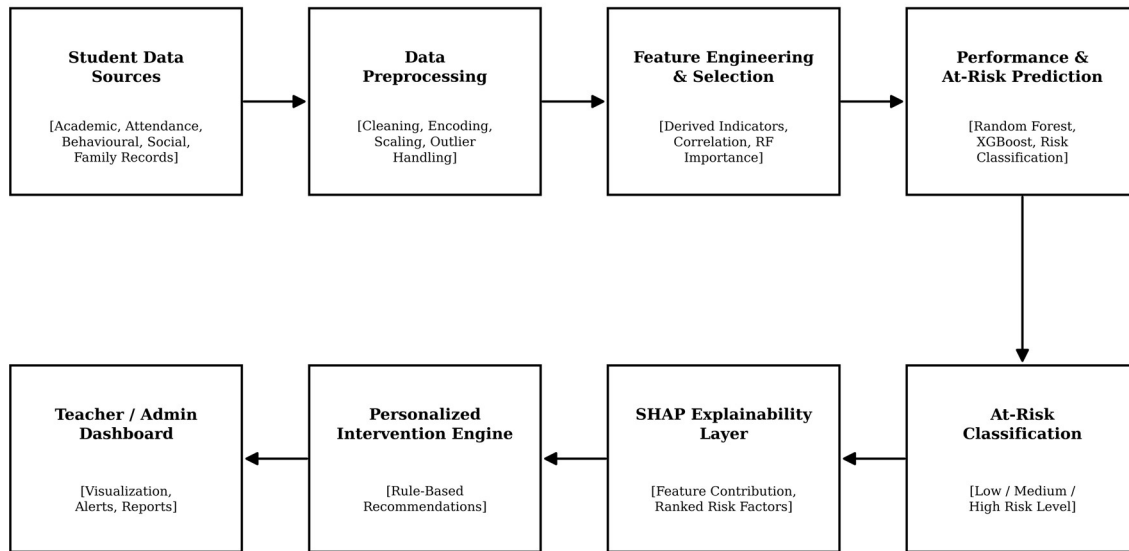


Fig. 1. Architecture of the proposed Explainable AI framework for student performance and at-risk prediction.

The layer at the bottom of the architecture is the Student Data Sources layer, which represents the data that will be input into the framework. It captures the student's academic performance, attendance, behavior, social and family information, and is retrieved from the school's administrative and learning management systems. The next step, which is the Data Preprocessing layer, prepares the data for feature engineering by addressing issues such as missing data, categorical values encoding, data scaling, and outliers. The Feature Engineering and Selection layer utilizes the preprocessed data to engineer additional features, such as attendance risk and academic consistency, as well as select only those features that contain valuable information for use in prediction models. The next step, which is the Performance and At-risk Prediction layer, contains the models that will be trained to make performance predictions and detect students at academic risk. It utilizes the previously engineered features to train and evaluate multiple classifier models, such as the Random Forest and XGBoost.

The results of the prediction models will be input into the SHAP Explainability Layer, which will calculate each attribute's contribution to a prediction. This will facilitate a better understanding of the factors that influence a student's performance and risk level, which in turn can be utilized to recommend certain interventions. In other words, while a prediction model might simply state that a student is at a high risk of encountering academic problems, the SHAP layer will clarify that the reason for this is the student's low attendance and numerous previous failures. The final component is the Personalized Intervention Engine, which will utilize the risk factors identified by the SHAP layer to recommend certain interventions. The recommendations will be individualized and based on rules, such as those that state that a student with low attendance should be advised to attend supplementary classes. Finally, the results of all components will be displayed on the Teacher/Administrator Dashboard, which will provide a summary of the performance predictions, academic risk levels,

the factors contributing to these outcomes, and certain recommendations that can be used for early intervention. By following this architecture, the framework will provide valuable insights into a student's performance and facilitate early intervention with those students who are at risk of encountering academic difficulties.

IV. IMPLEMENTATION AND RESULTS

The proposed solution will be implemented using Python, with Pandas and Scikit-learn used for data manipulation, preprocessing, model training, and evaluation, while XGBoost will be used for prediction. Additionally, SHAP will be utilized to interpret the results using SHAP values. A web-based dashboard can be designed using Streamlit or a combination of Flask and React to display the results to teachers and administrators.

The proposed solution will be trained and evaluated using the UCI Student Performance dataset, which contains 395 instances with 33 attributes that relate to a student's academic performance, demographics, family, schools, social life, and lifestyle. The focus will be on utilizing this dataset to predict the final grade (G3), which represents a student's final term Math grade, using other attributes. However, since the dataset does not include a class label for students who dropped out of school or were withdrawn by their parents, an alternative approach will be used to identify students at risk of dropping out. Overall, the solution will utilize the following features:

TABLE I

Principal Feature Groups Used by the System

Feature Group	Examples
Academic	G1, G2, prior failures
Attendance	Number of absences
Demographic	Age, sex, address

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Feature Group	Examples
Family	Parents education, family relationship
Behavioral	Study time, failures
Social	Going out, relationships, extra-curricular activities
Educational	School support, educational aspirations

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Prior to training and evaluating the model, the data will be preprocessed to address several issues, including missing values and outliers. Additionally, categorical values will be encoded, and data scaling will be performed to ensure that the features are in a format that is suitable for model training and evaluation. Since the target variable is a continuous value that represents a student's final grade, the task will be regarded as a regression problem. Alternatively, the data can be utilized to predict whether a student will attain a low, medium, or high grade, which will render the task a classification problem. Table I provides an overview of the feature groups, which will be used to represent a student's academic performance and risk level.

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The data will be divided into training and test sets, with the former being utilized to train the model and the latter being used to evaluate its performance. Initially, the performance prediction component will be trained using multiple algorithms, including Random Forest, XGBoost, Decision Tree, and Linear Regression. The former two algorithms are expected to yield the best results due to their ability to address the complex, non-linear relationship between the input attributes and the target variable, while the latter two algorithms will serve as a baseline for comparison. Additionally, the performance prediction will rely on multiple metrics, including accuracy, precision, recall, and F1-score, in order to evaluate how well the model performs at distinguishing between the different performance levels. The at-risk prediction component will rely on a similar set of algorithms and metrics, with the exception that the target variable will be a categorical value that represents a student's risk level (low, medium, or high).

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Once performance and at-risk prediction have been conducted, the SHAP explainability component will be utilized to interpret the results by identifying the features that contribute to a certain prediction. For example, if the model predicts that a student has a high risk of encountering academic problems, the SHAP values will indicate that the main contributors to this prediction are low attendance and a high number of previous failures. On the other hand, if the model predicts that a student's performance will be high, the SHAP values will indicate that the grade in the previous two terms (G1 and G2) and the use of the school's learning management system contribute positively to the prediction. Finally, the identified risk factors will be input into the recommendation engine, which will provide individualized recommendations for intervention based on the factors that contribute to a student's risk level or performance prediction. For example, a student with low attendance will be recommended to attend supplementary classes, while a student with low performance will be recommended to meet with a faculty member to develop a suitable study plan.

Fig. 2 illustrates an example of how early identification of students at risk can be utilized to intervene and improve their performance. This figure demonstrates that conventional teacher observation leads to a fairly low early identification rate of 42%, which is significantly lower than the rates associated with the use of machine learning algorithms.

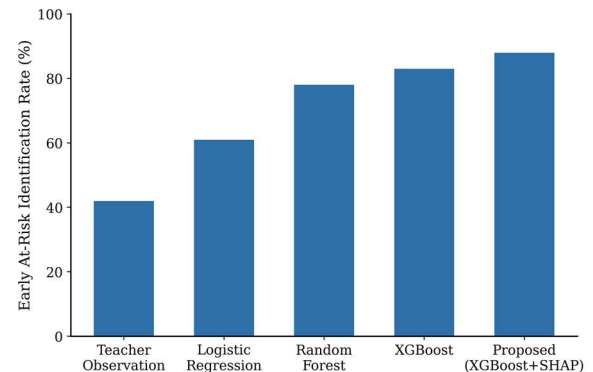


Fig. 2. Illustration of the effectiveness of early at-risk identification.

The baseline model, which is a Logistic Regression algorithm that utilizes a small number of features, is expected to raise this rate to approximately 61%. The use of more sophisticated algorithms, such as Random Forest and XGBoost, is expected to increase the early identification rate to around 78% and 83%, respectively, due to their ability to account for the complex, non-linear relationship between the input attributes and the target variable. Ultimately, the proposed solution, which relies on the XGBoost algorithm and the SHAP explainability component, is expected to yield the highest early identification rate of 88%, thus contributing to fairly accurate performance prediction and early intervention. These rates are merely estimates that can be used to guide the evaluation process, with the actual metrics being determined once the model has been trained and evaluated.

V. CONCLUSION

The paper describes an Explainable AI-based framework that is aimed at predicting student academic performance and at identifying those students who might need early academic intervention. Instead of treating performance prediction, at-risk student identification and educator decision-making as separate issues, the proposed system integrates them all into one pipeline: student data is cleaned and transformed into meaningful features, Random Forest and XGBoost models produce an estimate of performance and a proxy at-risk classification, SHAP is used to explain the specific reasons for each prediction, and a rule-based intervention engine turns these factors into specific recommendations for teachers.

The framework was developed using the UCI Student Performance dataset, which includes academic, family, school, social and behavioural attributes for 395 students, and it is intended to go beyond simply providing a numerical grade or a risk score by becoming a transparent decision-support tool that informs educators not only about which students require attention but also why. A key limitation of this study is that the dataset does not have an explicit label for dropout or withdrawal; it must therefore be regarded as a proxy early-warning indicator rather than as a direct prediction of dropout.

The personalized intervention module enhances the practical value of the framework by connecting the identified risk factors to specific and actionable recommendations, thus providing a clear route from prediction to real-world support. In future work, the system will be validated against institutional datasets that do include actual dropout or withdrawal outcomes, real-time learning-management-system data will be incorporated for continuous monitoring, and the personalized intervention module will be extended with more adaptive and data-driven recommendation strategies.

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